OncoInsights — Executive Summary

Cohort: TCGA Lung Adenocarcinoma (LUAD), PanCancer Atlas, cBioPortal | **n = 503 patients** (of 566 pulled; see Data Quality note)

Problem Framing

Clinical teams reviewing a lung adenocarcinoma cohort need a fast way to answer three questions: who is highest-risk, which mutations are driving that risk, and whether a simple risk score actually separates patients by outcome. OncoInsights builds a reproducible pipeline — from raw clinical/mutation/expression data through a validated composite risk score, SQL cohort views, statistical testing, and an interactive dashboard — to answer exactly those questions for this cohort, end to end.

Key Findings

1. **The engineered risk score meaningfully separates survival outcomes.** Patients flagged high-risk (top quartile of `RISK_SCORE`, n=126, 25% of cohort) show significantly worse overall survival than the rest of the cohort (log-rank p=0.0021; median survival 38.5 vs 53.7 months). This is not a foregone conclusion — a composite score with hand-picked weights could easily fail to separate outcomes, and here it doesn't.
2. **TP53 is the dominant mutation in this cohort (52% of patients)**, consistent with published TCGA-LUAD mutation rates. This cross-check against known literature is a useful sanity check that the mutation data pulled from cBioPortal is behaving as expected.
3. **Most driver-gene mutations shift their own gene's expression** — 7 of 8 tested genes (TP53, KRAS, EGFR, STK11, KEAP1, SMARCA4, NF1) show a statistically significant expression difference between mutated and wild-type patients after FDR correction (only ATM did not reach significance). This is a strong internal consistency check between the mutation calls and expression data.
4. **TP53 mutation status is not associated with tumor stage at diagnosis** (chi-square p>0.05 in Module 6) — despite being the most common mutation, it doesn't track with how advanced disease is at diagnosis, suggesting it's an early, stage-independent driver event rather than a late-stage marker.
5. **Mutation burden (within the 40-gene panel) has a weak, unexpected negative correlation with age** (r=-0.165, p=0.0002). This runs counter to the common intuition that mutation burden rises with age, and is flagged here as a finding worth follow-up rather than smoothed over.
6. **Early-stage disease dominates the cohort** (79% Stage I/II at diagnosis), which is favorable from a treatment-options standpoint but also means the cohort is not evenly powered across all four stages (Stage IV: only 27 patients) — a caveat for any stage-IV-specific finding.

7. **EGFR expression level alone is not a strong standalone prognostic signal** in this panel — mortality rate is roughly flat across EGFR expression quartiles (SQL Q11), meaning expression level shouldn't be over-interpreted without also considering mutation status and stage.

Risk Factors Identified

- **Higher driver-panel mutation burden** — direct input to the validated risk score.
- **Older age at diagnosis** — direct input to the risk score; also the single largest cohort segment (60-75 years) and worth prioritizing for risk-score-based triage.
- **Later AJCC stage (III/IV)** — direct input to the risk score, and the smallest but most vulnerable segment of the cohort.

Recommendations

1. **Use the composite risk score to prioritize clinical follow-up, not any single input in isolation.** The risk score's statistically validated survival separation (Finding 1) makes it a reasonable triage signal; none of its three individual inputs (mutation count, age, stage) showed anywhere near as clean a survival split on its own in Module 6.
2. **Invest in closing clinical follow-up data gaps before scaling this pipeline to a larger cohort.** 63 of 566 pulled patients (11%) were dropped during cleaning — 52 of those had no usable clinical record at all (every core field blank), and the rest were missing just stage or survival time specifically. Disease-free-survival fields were also missing for ~47% of the remaining cohort — that's a meaningful chunk of statistical power left on the table purely from incomplete data capture, not from the underlying biology.
3. **Don't treat single-gene expression (e.g. EGFR) as a standalone risk marker in this cohort.** Finding 7 shows expression level alone is a weak signal; mutation status and stage are doing more of the real stratification work, and dashboard consumers should be steered toward the composite risk score rather than single-gene cutoffs.

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